

* In our lesson I (for whatever reason) wrote $M = P/V$ instead of $\rho = M/V$.

$$* TdS = dE + \hat{P}dV$$

ρ : density

$\hat{P}$: Pressure

• Adiabatic $\Rightarrow dS = 0$, divide by dt :

$$0 = dE + \hat{P}dV$$

$$0 = \frac{dE}{dt} + \hat{P} \frac{dV}{dt} \quad \left| \text{Since } \rho = \frac{M}{V} \right.$$

$$• E = Mc^2 = (\rho V)c^2$$

$$• V = \frac{4\pi}{3}R^3$$

$$\text{I.e.) } \frac{dE}{dt} = \frac{d}{dt}(\rho V)c^2 = \dot{\rho} V c^2 + \rho \dot{V} c^2$$

Plugging this in:

$$\frac{dE}{dt} + \hat{P} \frac{dV}{dt} = (\dot{\rho} V c^2 + \rho \dot{V} c^2) + \hat{P} \dot{V} = 0$$

• Combine $\dot{V}$ terms

$$\dot{\rho} V c^2 + \dot{V}(\rho c^2 + \hat{P}) = 0$$

• Divide by V :

$$\dot{\rho} c^2 + \frac{\dot{V}}{V}(\rho c^2 + \hat{P}) = 0$$

$$• \text{Since } \dot{V} = 4\pi R^2 \dot{R} \Rightarrow \frac{\dot{V}}{V} = \frac{(4\pi R^2 \dot{R})}{(\frac{4\pi}{3} R^3)} = \frac{3\dot{R}}{R}$$

thus,

$$\dot{\rho} c^2 + \frac{3\dot{R}}{R}(\rho c^2 + \hat{P}) = 0$$

• Multiply by $1/c^2$:

$$\dot{\rho} + \frac{3\dot{R}}{R}(\rho + \frac{\hat{P}}{c^2}) = 0$$

